SECTION 09 9600 - HIGH-PERFORMANCE COATINGS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. High performance coatings for storage room as noted on the drawings.

B. Special preparation of surfaces.

C. Materials included in this section shall meet the requirements for High Performance Sustainable Buildings for New Construction points for the following Sections:

1. 2-5.3.2 - Reduce Volatile Organic Compounds (VOC) Low-Emitting Materials

2. The contractor is expected to understand the HPSB requirements for these sections and include all applicable overhead in their base bid for the necessary documentation to achieve the above listed credits.

1.02 RELATED REQUIREMENTS

A. Section 01 6116 - Volatile Organic Compound (VOC) Content Restrictions.

B. Section 09 9123 - Interior Painting

1.03 REFERENCE STANDARDS

A. ASTM E 84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2010.

B. SSPC-SP 2 - Hand Tool Cleaning; Society for Protective Coatings; 1982 (Ed. 2004).

C. SSPC-SP 3 - Power Tool Cleaning; Society for Protective Coatings; 1982 (Ed. 2004).

D. ***ASTM International (ASTM): (Added by Addendum No. 1)***

1. ***ASTM C 413 - Standard Test Method for Absorption of Chemical-Resistant Mortars, Grouts, Monolithic Surfacings, and Polymer Concretes.***

2. ***ASTM D 635 - Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position.***

3. ***ASTM D 695 - Standard Test Method for Compressive Properties of Rigid Plastics.***

4. ***ASTM D1475 - Standard Test Method For Density of Liquid Coatings, Inks, and Related Products.***

5. ***ASTM D 2047 - Standard Test Method for Static Coefficient of Friction of Polish-Coated Flooring Surfaces as Measured by the James Machine.***
6. ***ASTM D 2240 - Standard Test Method for Rubber Property—Durometer Hardness.***
7. ***ASTM D 2244 - Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates.***
8. ***ASTM D2369 - Standard Test Method for Volatile Content of Coatings.***
9. ***ASTM D 2370 - Standard Test Method for Tensile Properties of Organic Coatings.***
10. ***ASTM D 3960 - Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings.***
11. ***ASTM D 4060 - Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser.***
12. ***ASTM D 4366 - Standard Test Methods for Hardness of Organic Coatings by Pendulum Damping Tests***
13. ***ASTM D5441 - Standard Test Method for Analysis of Methyl Tert-Butyl Ether (MTBE) by Gas Chromatography.09670-2***
14. ***ASTM D 7234 - Standard Test Method for Pull-Off Adhesion Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers.***
15. ***ASTM F 1869 - Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride.***
16. ***ASTM F 2170 - Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes***
17. ***ASTM G 154 - Standard Practice for Operating Fluorescent Ultraviolet (UV) Lamp Apparatus for Exposure of Nonmetallic Materials.***

E. ***International Concrete Repair Institute (ICRI): (Added by Addendum No. 1)***

1. ***ICRI 310.2R - Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, Polymer Overlays, and Concrete Repair.***

F. ***Military Specifications (MIL): (Added by Addendum No. 1)***

1. ***MIL-D-3134J - Deck Covering Materials.***

G. ***National Floor Safety Institute (NFSI): (Added by Addendum No. 1)***

1. ***ANSI/NFSI B101.1 - Test Method for Measuring Wet SCOF of Common Hard-Surface Floor Materials.***

1.04 SUBMITTALS

A. See Section 01 3001 - Submittals, for submittal procedures

B. ~~Product Data: Provide data indicating coating materials.~~ ***(Deleted by Addendum No. 1)***

C. ***Product Data: (Added by Addendum No. 1)***

1. ***Manufacturer's data sheets on each product to be used, including properites, VOC content, wet static coefficient of friction, compressive strength, tensile strength, eloongation and similar properties.***
2. ***Preparation instructions and recommendations.***
3. ***Storage and handling requirements and recommendations.***
4. ***Typical installation methods***

D. ***Verification Samples: Two representative units of each system, including color and texture (Added by Addendum No. 1)***

E. Manufacturer's Installation Instructions: Indicate special procedures, perimeter conditions requiring special attention, and recommended cleaning procedures for preparing existing concrete floor slabs to receive new coating. .

F. Maintenance Data: Include cleaning procedures and repair and patching techniques.

G. Maintenance Materials: Furnish the following for the Government's use in maintenance of project.

1. Extra Coating Materials: 1 gallon (4 liters) of each type and color.
2. Label each container with manufacturer's name, product number, color number, and room names and numbers where used.

H. Sustainability Submittals, Product data for HPSB Compliance:

1. For products containing VOCs, documentation (material safety data sheets (MSDS), third-party certificates, or test reports) showing printed statement of VOC content.

2. VOC Content Limitations: For the specified products, submit documentation of conformance with Specification Section 01 6116 - Volatile Organic Compound (VOC) Content Restrictions.

1.05 QUALITY ASSURANCE

A. Maintain one copy of each referenced document that applies to application on site.

B. Manufacturer Qualifications: Company specializing in manufacturing the Products specified in this section with minimum three years documented experience.

C. Applicator Qualifications: Company specializing in performing the work of this section with minimum 3 years documented experience.

D. ***Source Limitations: Provide each type of product from a single manufacturing source to ensure uniformity (Added by Addendum No. 1)***

1.06 MOCK-UP

A. Provide mock-up, 10 feet long by 10 feet wide, illustrating coating , for each specified coating.

B. Locate where directed.

C. Mock-up may remain as part of the Work.

1.07 FIELD CONDITIONS

A. Do not install materials when temperature is below 55 degrees F (13 degrees C) or above 90 degrees F (32 degrees C).

B. Maintain this temperature range, 24 hours before, during, and 72 hours after installation of coating.

C. Provide lighting level of 80 ft candles (860 lx) measured mid-height at substrate surface.

D. Restrict traffic from area where coating is being applied or is curing.

1.08 WARRANTY

A. See Section 01 7800 - Closeout Submittals, for additional warranty requirements.

B. Correct defective Work within a five year period after Date of Beneficial Occupancy.

C. Warranty: Include coverage for bond to substrate and degradation of chemical resistance.

PART 2 PRODUCTS

2.01 ~~HIGH-PERFORMANCE COATINGS~~ (Deleted by Addendum No. 1)

A. ~~Provide coating systems that meet the following minimum performance criteria, unless more stringent criteria are specified:~~

1. ~~Abrasion Resistance: 18 mg loss, when tested in accordance with ASTM D4060.~~

B. ~~Moderate Exposure: All minimum criteria, plus:~~

2.02 ~~MATERIALS~~ (Deleted by Addendum No. 1)

A. ~~Coatings - General: Provide complete multi-coat systems formulated and recommended by manufacturer for the applications indicated, in the thicknesses indicated; number of coats specified does not include primer or filler coat.~~

B. ~~Polyurethane Floor Coating: Three coats, moisture cure polyurethane, non-skid finish.~~

1. ~~Product characteristics:~~

a. ~~Percentage of solids by weight (ASTM D2369): 99.35 - Part A; 59.23 - Part B, 100 - Part C, minimum.~~

b. ~~Dry film thickness, per coat: 3.0, minimum.~~

c. ~~VOC Content per ASTM D3960 (Mixed): 86 g/L maximim A+B+C~~

d. ~~Abrasion Resistance: mg loss ASMT D4060 = 18.0~~

e. ~~Coefficient of Friction ASTM D2047 = 0.63~~

f. ~~Tension Strength (psi) Resin Only - ASTM C2370= 6,250~~

2. ~~Chemical Resistance Properties:~~

a. ~~Product shall have excellent resistant (in both 1 day and 7 day tests) to the following materials:~~

1) ~~Acids (Inorganic): 10% Hydrochloric Acid, 30% Hydrochloric Acid, 10% Nitric Acid, 37% Sulfuric Acid,~~

2) ~~Acids (Organic): 10% Acetic Acid, 10% Citric Acid, Oleic Acid,~~

3) ~~Alkalies: 10% Ammonium Hydroxide, 50% Sodium Hydroxide,~~

4) ~~Solvents(Alcohols): Ethylene Glycol (Antifreeze), Isopropyl Alcohol, Methanol~~

5) ~~Solvents (Aliphatic:) d-Limonene, Jet Fuel (JP-4), Gasoline, Mineral Spirits~~

6) ~~Solvents (Aromatic) Xylene~~

7) ~~Solvents (Ketones & Esters): Methyl Ethyl Ketone, Propylene Glycol Methyl, Ether Acetate~~

8) ~~Miscellaneous: 20% Ammonium Nitrate, Chemicals Brake Fluid, Bleach, Motor Oil (SAE30), Skydrol® 500B, Skydrol® LD4, 20% Sodium Chloride, 1% Tide® Laundry Soap, 10% Trisodium Phosphate~~

3. ~~Primer for concrete: Two component epoxy primer as recommended by the coating manufacturer.~~

4. ~~Colorant: Provide manufacturer's standard colorant. Color as selected by the Government from manufacturer's full line of available colors.~~

5. ~~Traction Grit: Provide aluminum oxide grit materials as recommended by the manufacturer.~~

6. ~~Cleaners & Related Products: Provide Industrial Grease Remover, detergents, cleaners, and etchants as recommended by the manufacturer for applications indicated on the contract documents.~~

2.03 *COATING SYSTEM (Added by Addendum No. 1)*

A. ***Provide Coating System Containing the Following Components:***

1. ***Primer Coat (3-5 mils thickness)***

2. ***Base Coat (17-19 mils thickness)***

3. ***Topcoat (3 mil minimum dry thickness)***

B. ***System Properties***

1. ***Abrasion Resistance Taber Abraser CS-17 Taber Abrasion Wheel, 1,000 gram load, 1,000 revolutions, ASTM D4060, 18 mg/loss***

2. ***Adhesion to Concrete, psi (MPa), ASTM D4541, 450 [3.10] (concrete failed)***

3. ***Adhesion to concrete, psi [MPa], ASTM D7234, 732[4.48] (concrete failed)***
4. ***Coefficient of Friction-COF, James Friction Tester, ASTM D2047, 0.63***
5. ***Coefficient of Friction-Wet Static, BOT 3000, ANSI/NFSI B101.1, 0.94***
6. ***Compressive Strength, psi [MPa], ASTM D695, 13,500 [93.08]***
7. ***Flammability, mm/min, ASTM D635, 182***
8. ***König Hardness 93 mil/0.08 mm film), ASTM D4366, 171.3***
9. ***Resistance to Yellowing as measured using ASTM D2244 after 1000 consecutive hours UV exposure in QUV, ASTM G154, <10 increase of yellow units (CIE Lab Δ) if pigmented topcoat***
10. ***Shore D Hardness, ASTM D2240, 80-85 @ 0 sec|75-80 @ 15 sec***
11. ***Tensile Strength, psi [MPa], ASTM D2370, 6,250 [43.09]***
12. ***Percent Elongation, ASTM D2370, 6%***
13. ***Volatile Organic Compound, VOC lb/gal [g/l], ASTM D3960:***
    a. ***Primer/Base Coat: A+B=0.14 [49]***
    b. ***Top Coat A+B=0.05 [6]***
14. ***Water Absorption, 24- hour immersion, ASTM C413, 0.2% weight increase***

2.04 ***PRODUCT PROPERTIES (Added by Addendum No. 1)***

A. ***Primer/Base Coat:***

1. ***Percent Solids, by weight (by volume), ASTM D1475, A + B: 95.45 (94.56).***
2. ***Volatile Organic Compound-VOC, ASTM D3960, Mixed A + B: 0.41 lb./gal (49 g/L).09670-5***
3. ***Abrasion Resistance, mg loss, Taber Abraser, C-17 Taber Abrasion Wheel, 1,000 gram load, 1,000 revolutions, ASTM D4060: 83.1.***
4. ***Coefficient of Friction-COF, James Friction Tester, ASTM D2047: 0.59-0.62.***
5. ***Adhesion to Concrete, ASTM D5441: 732 psi (4.48 MPa) concrete failed.***
6. ***Adhesion to Concrete, ASTM D7234: 450 psi (3.10 MPa) concrete failed.***

7. ***Compressive Strength, ASTM D695: 13,500 psi (93.079 MPa).***

8. ***Tensile Strength, ASTM D2370: 8,000 psi (55.158 MPa).***

9. ***Percent Elongation, ASTM D2370: 5.***

10. ***Shore D Hardness, ASTM D2240: 80-85 at 0 sec, 75-80 at 15 sec.***

B. ***Top Coat: A clear high solids, three-component, satin finish, aliphatic, moisture-cure urethane.***

1. ***Percent Solids, by weight (by volume), ASTM D2369, A + B + C: 94.02 (92.57)***

2. ***Volatile Organic Compound-VOC, ASTM D3960, Mixed A + B + C: 0.05 lb/gal (6 g/L).***

3. ***Abrasion Resistance, mg loss, Taber Abraser, C-17 Taber Abrasion Wheel, 1,000 gram load, 1,000 revolutions, ASTM D4060: 18.***

4. ***Coefficient of Friction-COF, James Friction Tester, ASTM D2047: 0.63.***

5. ***Wet Static Coefficient of Friction, BOT 3000, ANSI/NFSI B101.1: 0.94.***

6. ***Flammability, ASTM G154: 182 mm/min.***

7. ***Resistance to Yellowing as measured using ASTM D2244 after 1000 consecutive hours UV exposure in QUV, ASTM G154, Less than 10 increase of yellow units (CIE Lab Δb)***

8. ***Tensile Strength, (resin only), ASTM D2370: 6,250 psi (43,092 MPa).***

9. ***Percent Elongation, (resin only), ASTM D2370: 6.***

10. ***König Hardness, (3 mil/76.2 micron film), ASTM D4366: 171.3.***

11. ***Water Absorption, 24-hour immersion, ASTM C413: 0.2 percent weight increase.***

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify existing conditions before starting work.

B. Verify that substrate surfaces are ready to receive work as instructed by the coating manufacturer. Obtain and follow manufacturer's instructions for examination and testing of substrates.

C. Examine concrete surface to receive floor coating system. Notify the Contracting Officer if surface is not acceptable. Do not begin surface preparation or application until unacceptable conditions have been corrected. Proceeding with installation of coating system shall be an indication that floor substrate has been review and approved by applicator.

D. Allow concrete substrate to cure a minimum of 30 days.

E. CHECK THE TEMPERATURE AND HUMIDITY: Floor temperature and materials should be between 65°F (18°C) and 90°F (32°C). Humidity must be less than 80%. DO NOT coat unless floor temperature is more than five degrees over the dew point.

F. CHECK FOR MOISTURE: Concrete must be dry before application of this floor coating material. Concrete moisture testing must occur. Calcium chloride testing or in-situ relative humidity testing is recommended. Readings must be below 3 pounds per 1,000 square feet over a 24-hour period on the calcium chloride test or below 70% relative internal concretehumidity. Test methods can be purchased at www.astm.org, see ASTM F1869 or F2170, respectively or follow instructions from the suppliers of these tests.

3.02 PREPARATION

A. Clean surfaces of loose foreign matter.

B. Remove substances that would bleed through finished coatings. If unremovable, seal surface with shellac.

C. Remove finish hardware, fixture covers, and accessories and store.

D. Existing Painted and Sealed Surfaces:

1. Remove loose, flaking, and peeling paint. Feather edge and sand smooth edges of chipped paint.

2. Clean with mixture of trisodium phosphate and water to remove surface grease and foreign matter.

E. Protect adjacent surfaces and materials not receiving coating from spatter and overspray; mask if necessary to provide adequate protection. Repair damage.

3.03 PRIMING

A. Concrete: Prior to priming, patch with masonry filler to produce smooth surface.

B. Prepare surface in accordance with manufacturer's instructions.

1. Cleaning: Scrub with Tennant detergent and rinse with clean water to remove surface dirt, grease and oil.

2. Removing: Remove coatings and curing membranes with one of the following methods:

    a. Mechanical - Sand floors.

    b. Chemical - Use chemicals as recommended by manufacturer.

    c. Conditioning:

        1) Apply conditioning as recommended by manufacturer with manufacturer's recommended conditioning products.

        2) Do not use unbuffered muriatic acid to condition the concrete.

3.04 COATING APPLICATION

A. Apply coatings in accordance with manufacturer's instructions, to thicknesses specified.

1. Assemble squeegees and rollers; clean rollers to remove residual lint.

    a. Apply primer approved by manufacturer

        1) Mix components together.

        2) Mix only enough material which can be applied within 20 minutes.

        3) Apply primer at the rate of 321-535 ft2/gal.

        4) Allow primer to cure 8 hours at 75 degrees F (24 degrees C) and 50% relative humidity.

        5) Sand with 80 grit sandpaper, detergent scrub and rinse with clean water to ensure adequate adhesion between coats.

2. Coating: Apply coating in accordance with all manufacturer's written instructions.

    a. Open and mix only enough material which can be applied in a 2 hour period.

        1) Apply coating at the rate of 500 ft2/gal.

        2) Allow coating to dry 24 hours at 75 degrees F (24 degrees C) and 50% relative humidity.

3. Broadcast traction grit in accordance with manufacturer's written instructions. Apply at a rate of 2 lbs per 1,000 square feet to achieve medium grit pattern.

B. Apply in uniform thickness coats, without runs, drips, pinholes, brush marks, or variations in color, texture, or finish. Finish edges, crevices, corners, and other changes in dimension with full coating thickness.

3.05 CLEANING

A. Collect waste material that could constitute a fire hazard, place in closed metal containers, and remove daily from site.

B. Clean surfaces immediately of overspray, splatter, and excess material.

C. After coating has cured, clean and replace finish hardware, fixtures, and fittings previously removed.

3.06 PROTECTION

A. Close job site to traffic for a period of 24 hours after coating application

B. Protect in place coatings from damage from adjacent construction activities in accordance with manufacturer’s instructions.

C. Repair any damage to coating caused by adjacent construction activities prior to Beneficial Occupancy.

D. Clean floor coating in accordance with division one section specifications and manufacturer’s written instructions.

END OF SECTION